

DAY - 03: Bloch Sphere & State Representation

Comprehensive Technical Study Notes

1. The Necessity of State Representation

In a classical architecture, database tracking is strictly deterministic and binary: a bit resolves uniquely to either **0** or **1**. Conversely, a quantum system leverages a continuous state space enabling a qubit to inhabit an infinite number of linear combinations of these baseline states. To compute with such systems, we utilize **Quantum State Representation** to mathematically establish:

- **Probability Distribution:** Exactly how much of state $|0\rangle$ and state $|1\rangle$ exist simultaneously.
- **Phase Configurations:** The relative quantum phase relationship connecting the underlying amplitudes, which drives critical interference properties.

2. Mathematical State Vector Framework

A standalone pure quantum qubit is mathematically formalized as a state vector, designated by the convention $|\psi\rangle$ (Ket Psi). It expands as a linear combination of the computational basis vectors multiplied by complex numbers known as **probability amplitudes**:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where:

- **α (Alpha):** The complex probability amplitude mapping to the ground state $|0\rangle$.
- **β (Beta):** The complex probability amplitude mapping to the excited state $|1\rangle$.

Matrix Vector Basis Notation

The core computational standard basis choices are represented as orthogonal column matrices:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The Normalization Condition (The Born Rule)

Because physical measurement forces the qubit to collapse entirely into one of the two computational pathways without a third alternative, the total integrated probabilities must sum to exactly **1** (or 100%):

$$|\alpha|^2 + |\beta|^2 = 1$$

- $|\alpha|^2$ represents the precise probability of registering a **0** classical outcome.
- $|\beta|^2$ represents the precise probability of registering a **1** classical outcome.

3. Essential Quantum State Classifications

State A: Pure Ground State ($|0\rangle$)

- **Vector representation:** $|\psi\rangle = 1|0\rangle + 0|1\rangle = |0\rangle$
- **Statistical Breakdown:** Probability($|0\rangle$) = 100%, Probability($|1\rangle$) = 0%

State B: Pure Excited State ($|1\rangle$)

- **Vector representation:** $|\psi\rangle = 0|0\rangle + 1|1\rangle = |1\rangle$
- **Statistical Breakdown:** Probability($|0\rangle$) = 0%, Probability($|1\rangle$) = 100%

State C: Equal Superposition State ($|+\rangle$)

- **Vector representation:** $|\psi\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$
- **Statistical Breakdown:** Probability($|0\rangle$) = 50%, Probability($|1\rangle$) = 50%
- **Significance:** A bedrock state required to facilitate parallel processing workflows across algorithms.

4. The Bloch Sphere Visual Mapping Architecture

The **Bloch Sphere** functions as a geometric model providing a 3D visualization of a single isolated qubit. Every pure quantum state corresponds precisely to a single point matching coordinate parameters across the sphere's surface.

The Three Geometrical Basis Coordinates

Basis Axis	Primary Vector States	Physical Geometry Locations	Operational Role
Z-Basis	$ 0\rangle$ and $ 1\rangle$	North & South Poles	Standard computational basis states mapped during read/measurement operations.
X-Basis	$ +\rangle$ and $ -\rangle$	Equatorial Left & Right Surfacing	Equal balance superposition spaces foundational to algorithm scaling.
Y-Basis	$ +i\rangle$ and $ -i\rangle$	Equatorial Front & Back Spherical Points	Introduces complex number configurations and phase metrics (i).

Global Spherical Coordinate Transformation

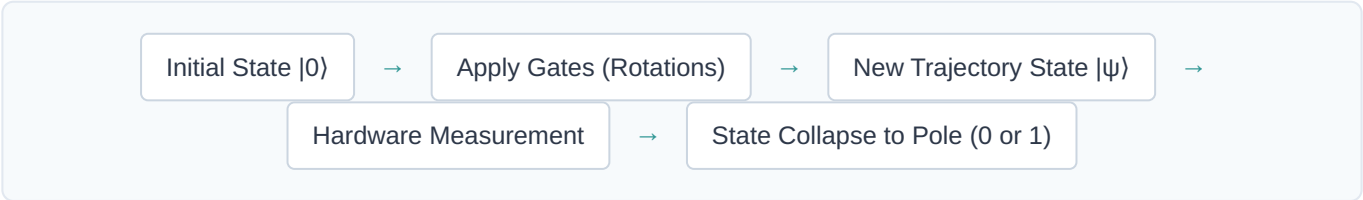
Using structural angular notation, any valid state can be written on the sphere as:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$$

- **Angle θ (Theta):** Dictates the structural latitude angle, adjusting the raw weight splitting between baseline states.
- **Angle φ (Phi):** Dictates the structural longitude angle, governing the phase rotation properties.

5. Pipeline Processing Flow & System Rotations

Unlike switching logic gates on classical transistors, quantum operations are physically executed by performing geometric **rotations of the state vector arrow along the Bloch Sphere** via quantum gates.



Architectural Limits of the Bloch Sphere Model

- **Strictly Limited Scope:** The Bloch Sphere model maps exactly **one qubit (1 Qubit)**.

- **The Entanglement Barrier:** When two or more qubits enter an **entangled state**, their information dependencies consolidate into an indivisible joint structure. The independent subsystems can no longer be modeled individually via standalone Bloch Spheres, requiring multi-dimensional state vectors.

Verified IBM Quantum Learning Materials:

- Core Bloch Sphere Module: [IBM Cloud Density Matrices Course Link](#)
- Architectural Foundations: [IBM Bits, Gates, and Circuits Course Link](#)